

associated resistance and capacitance of the generator. This integration network produces an output rise time equal to $2.2 RC$. This limitation is derived from the universal time-constant curve of a capacitor.

Figure 1-6 shows the effect of internal source resistance and capacitance on the equivalent circuit. At no time can the output risetime be faster than $2.2 RC$ or 2.2 nSec .

If a typical probe is used to measure this signal, the probe's specified input capacitance and resistance is added to the circuit as shown in Figure 1-7.

Because the probe's $10 \text{ M}\Omega$ resistance is much greater than the generator's 50 ohm output resistance, it can be ignored.

Figure 1-8 shows the equivalent circuit of the generator and probe, applying the $2.2 RC$ formula again. The actual risetime has slowed from 2.2 n Sec. to 3.4 nSec.

Percentage change in risetime due to the added probe tip capacitance:

$$\% \text{ change} = \frac{tr_2 - tr_1}{tr_1} \times 100 = \frac{3.4 - 2.2}{2.2} \times 100 = 55\%$$

Another way of estimating the affect of probe tip capacitance on a source is to take the ratio of probe tip capacitance (marked on the probe compensation box) to the known or estimated source capacitance.

Using the same values:

$$\frac{C_{\text{probe tip}}}{C_1} \times 100 = \frac{11 \text{ pF}}{20 \text{ pF}} \times 100 = 55\%$$

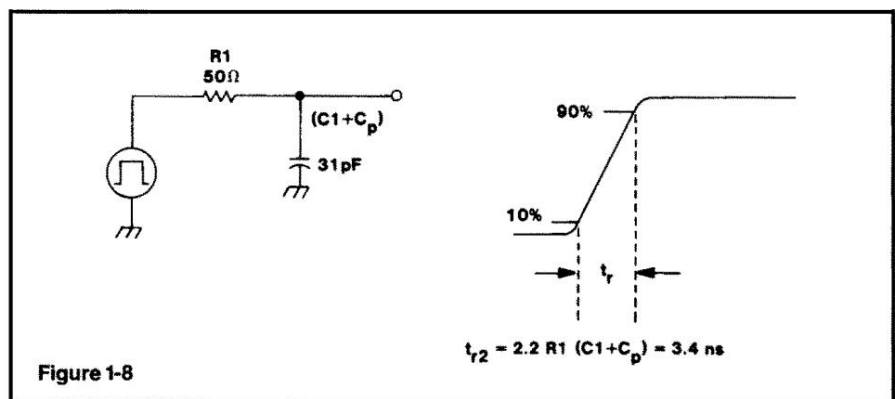
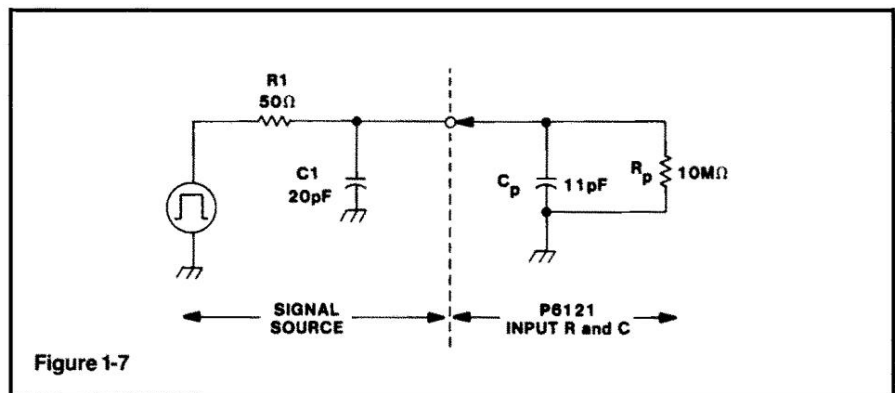
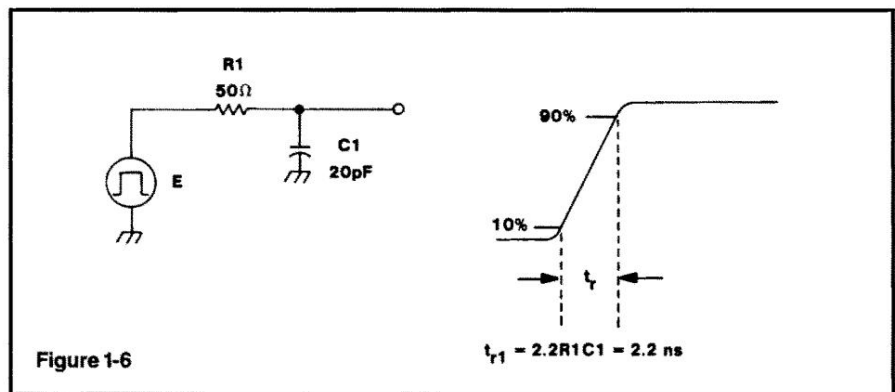
To summarize, any added capacitance slows the source rise-time when using high impedance passive probes. In general, the greater the attenuation ratio, the lower the tip capacitance. Here are some examples:

Probe	Attenuation	Tip Capacitance
Tektronix P6101A	X1	54 pF
Tektronix P6105A	X10	11.2 pF
Tektronix P6007	X100	2 pF

Capacitive Loading: Sinewave.

When probing continuous wave (CW) signals, the probe's capacitive reactance at the operating frequency must be taken into account.

The total impedance, as seen at the probe tip, is designated R_p and is a function of frequency. In addition to the capacitive and resistive elements, designed-in inductive elements serve to offset the pure capacitive loading to some degree.



Curves showing typical input impedance vs frequency, or typical X_p and R_p vs frequency are included in most Tektronix probe instruction manuals. Figure 1-9A shows the typical input impedance and phase relationship vs frequency of the Tektronix P6203 Active Probe. Note that the $10 \text{ K}\Omega$ input impedance is maintained to almost 10 MHz by careful design of the associated resistive, capacitive and inductive elements.

Figure 1-9B shows a plot of X_p and R_p vs frequency for a typical $10 \text{ M}\Omega$ passive probe. The dotted line (X_p) shows capacitive reactance vs frequency. The total loading is again offset by careful design of the associated R , C and L elements.

If you do not have ready access to the information and need a worst-case guide to probe loading, use the following formula:

$$X_p = \frac{1}{2\pi FC}$$

X_p = Capacitive reactance (ohms)
 F = Operating frequency
 C = Probe tip capacitance (marked on the probe body or compensation box.)

For example, a standard passive $10 \text{ M}\Omega$ probe with a tip capacitance of 11 pF will have a capacitive reactance (X_p) of about 290 ohm at 50 MHz .

Depending, of course, on the source impedance, this loading could have a major effect on the

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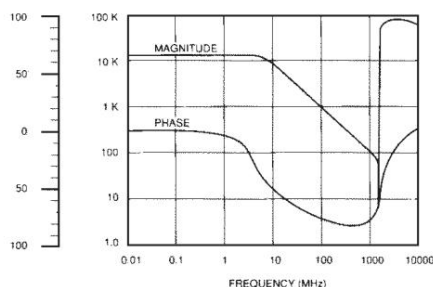


Figure 1-9A. Typical Input Impedance vs Frequency for the Tektronix P6203 Active Probe

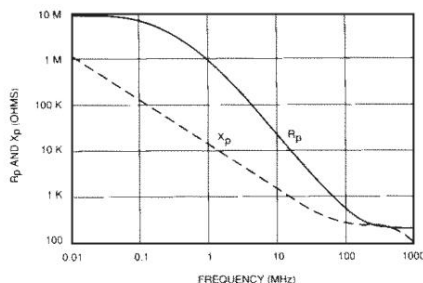


Figure 1-9B. X_p and R_p vs Frequency for a Typical 10 MΩ Passive Probe

signal amplitude (by simple divider action), and even on the operation of the circuit itself.

Resistive Loading. For all practical purposes, a 10X, 10M ohm passive probe has little effect on today's circuitry in terms of resistive loading, however, they do carry a trade-off in terms of relatively high capacitive loading as we have previously discussed.

Low Z Passive Probes. A "Low Z" passive probe offers very low tip capacitance at the expense of relatively high resistive loading. A typical 10X "50 ohm" probe has an input C of about 1 pF and a resistive loading of 500 ohm: Figure 1-10 shows the circuit and equivalent model of this type of probe.

This configuration forms a high frequency 10X voltage divider because, from transmission line theory, all that the 450 ohm tip resistor "sees" looking into the cable is a pure 50 ohm resistance, no C or L component. No low frequency compensation is necessary because it is not a capacitive divider. Low Z probes are typically high bandwidth (up to 3.5GHz and risetimes to 100 pS) and are best suited for making rise-time and transit-time measurements. They can, however, affect the pulse amplitude by simple resistive divider action between the source and the load (probe). Because of its resistive loading effects, this type of probe

performs best on 50 ohm or lower impedance circuits under test.

Note also that these probes operate into 50 ohm scope inputs only. They are typically teamed up with fast (500MHz to 1GHz) real time scopes or with scopes employing the sampling principle.

Bias-Offset Probes. A Bias/Offset probe is a special kind of Low Z design with the capability of providing a variable bias or offset voltage at the probe tip.

Bias/Offset probes like the Tektronix P6230 or P6231 are useful for probing high speed ECL circuitry, where resistive loading could upset the operating point. These special probes are fully described in Part 3; under Advanced Probing Techniques.

The Best of Both Worlds. From the foregoing, it can be seen that the totally "non-invasive" probe does not exist. However, one type of probe comes close—the active probe.

Active probes are discussed in the Tutorial section, but in general, they provide low resistance loading (10M ohm) with very low capacitive loading (1 to 2 pF). They do have trade-offs in terms of limited dynamic range, but under the right conditions, do indeed offer the best of both worlds.

Bandwidth. Bandwidth is the point on an amplitude versus frequency curve where the measurement system is down 3dB from a starting (reference) level. Figure 1-11 shows a typical response curve of an oscilloscope system.

Scope vertical amplifiers are designed for a Gaussian roll-off at the high end (a discussion of Gaussian response is beyond the scope of this primer). With this type of response, risetime is approximately related to bandwidth by the following equation:

$$Tr = \frac{.35}{BW} \quad \text{or, for convenience:} \quad \text{Risetime (nanoseconds)} = \frac{350}{\text{Bandwidth (MHz)}}$$

It is important to note that the measurement system is -3dB (30%) down in amplitude at the specified bandwidth limit.

Figure 1-12 shows an expanded portion of the -3dB area. The horizontal scale shows the input frequency derating factor necessary to obtain accuracies better than 30% for a specific bandwidth scope. For example, with no derating, a "100MHz" scope will have up to a 30% amplitude error at 100MHz (1.0 on the graph). If this scope is to have an amplitude accuracy better than 3%, the input frequency must be limited to about 30MHz (100MHz X .3).

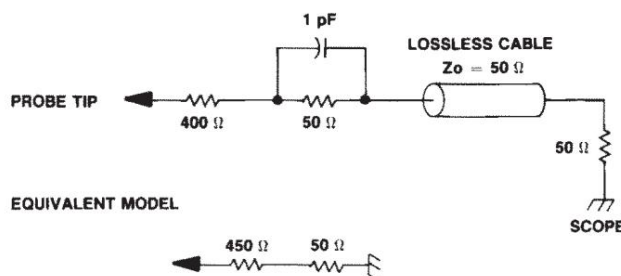


Figure 1-10

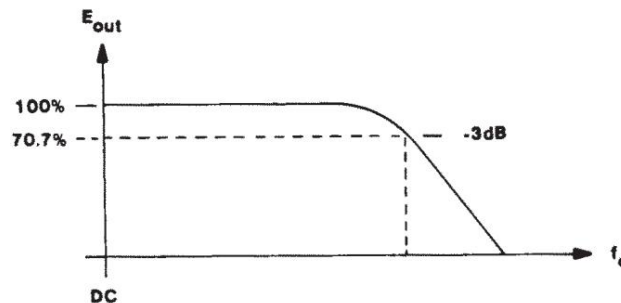


Figure 1-11

For making amplitude measurements within 3% at a specific frequency, choose a scope with at least four times the specified bandwidth as a general rule of thumb.

Probe Bandwidth. All probes are ranked by bandwidth. In this respect, they are like scopes or other amplifiers that are ranked by bandwidth. In these cases we apply the square root of the sum of the squares formula to obtain the "system risetime." This formula states that:

$$\text{Risetime system} = \sqrt{\text{Tr}_2^2 \text{ displayed} + \text{Tr}_2^2 \text{ source}}$$

Passive probes do not follow this rule and should not be included in the square root of the sum of the squares formula.

Tektronix provides a probe bandwidth ranking system that specifies "the bandwidth (frequency range) in which the probe performs within its specified limits. These limits include: total aberrations, risetime and swept bandwidth."

Both the source and the measurement system shall be specified when checking probe specifications (see Test Methods, this page).

In general, a Tektronix "100MHz" probe provides 100MHz performance (–3dB) when used on a compatible 100MHz scope. In other words, it provides full scope bandwidth **at the probe tip.**

However, not all probe/scope systems can follow this general rule. Refer to the sidebar, "Scope Bandwidth at the Probe Tip?"

Figure 1-13 shows examples of Tektronix scopes and their recommended passive probes.

Test Methods: As with all specifications, matching test methods must be employed to obtain specified performance. In the case of bandwidth and risetime measurements, it is essential to connect the probe to a properly terminated source. Tektronix specifies a 50 ohm source terminated in 50 ohm, making this a 25 ohm source impedance. Furthermore, the probe must be connected to the source via a proper probe tip to BNC adaptor. (Figure 1-14).

Figure 1-14 shows an equivalent circuit of a typical setup. The displayed risetime should be a 3.5 nSec or faster.

Figure 1-15 shows an equivalent circuit of a typical passive probe connected to a source.

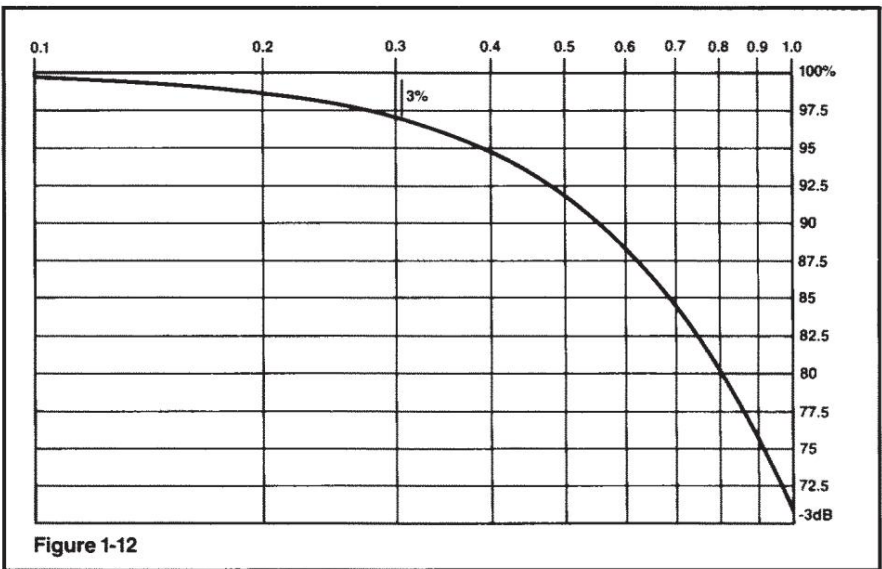


Figure 1-12

SCOPE	BW (1 M Ω input)	PROBE	BW	SYSTEM
2235	100	P6109	150	100
2245A	100	P6109	150	100
2246A	100	P6109	150	100
2445B	150	P6133	150	150
		Opt 25		
485	350	P6106A	250	250
2465B	400	P6137	400	400
2467B	400	P6137	400	400

Figure 1-13

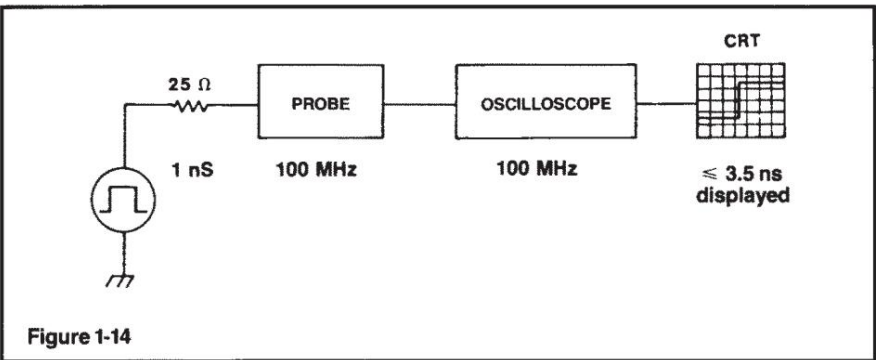


Figure 1-14

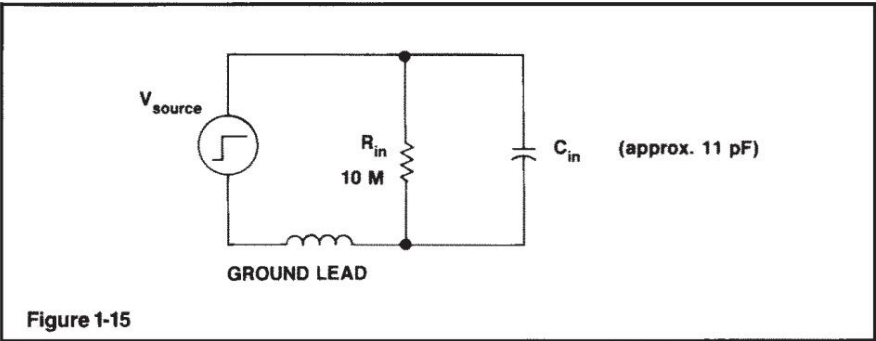


Figure 1-15

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Scope Bandwidth at the Probe Tip ?

Most manufacturers of general-purpose oscilloscopes that include standard accessory probes in the package, promise and deliver the advertised scope bandwidth **at the probe tip**.

For example, the Tektronix 2465B 400 MHz Portable Oscilloscope and its standard accessory P6137 Passive Probes deliver 400 MHz (-3db) at the probe tip.

However, not all high performance scopes can offer this feature, even when used with their recommended passive probes. For example, the Tektronix 11A32 400 MHz plug-in has a system bandwidth of 300 MHz when used with its recommended P6134 passive probe. This is simply because even the highest impedance passive probes are limited to about 300 to 350 MHz, while still meeting their other specifications.

It is important to note that the above performance is only obtainable under strictly controlled, and industry recognized conditions; which states that the signal must originate from a $50\ \Omega$ back-terminated source ($25\ \Omega$), and that the probe must be connected to the source by means of a probe tip to BNC (for other) adaptor.

This method ensures the shortest ground path and necessary low impedance to drive the probe's input capacitance, and to provide the specified bandwidth at the signal acquisition point, the probe tip.

Real-world signals rarely originate from $25\ \Omega$ sources, so less than optimum transient response and bandwidth should be expected when measuring higher impedance circuits.

How ground leads affect measurements

A ground lead is a wire that provides a local ground-return path when you are measuring any signal. An inadequate ground lead (one that is too long or too high in inductance) can reduce the fidelity of the high frequency portion of the displayed signal.

What grounding system to use.

When making **any** measurement, some form of ground path is required to make a basic two-terminal connection to the DUT. If you want to check the presence or absence of signals from low-frequency equipment, **and** if the equipment is line-powered and plugged into the same outlet system as the scope, then the common 3-wire ground system provides the signal ground return. However, this indirect route adds inductance in the signal path—it can also produce ringing and noise on the displayed signal and is not recommended.

When making any kind of absolute measurement, such as amplitude, risetime or time delay measurements, you should use the shortest grounding path possible, consistent with the need to move the probe among adjacent test points. The ultimate grounding system is an in-circuit ECB (etched circuit board) to probe tip adaptor. Tektronix can supply these for either miniature, compact or subminiature probe configurations.

Figure 1-15 shows an equivalent circuit of a typical passive probe connected to a source. The ground lead L and C_{in} form a series resonant circuit with only $10\text{M}\ \Omega$ for damping. When hit with a pulse, it will ring. Also, excessive L in the ground lead will limit the changing current to C_{in} , limiting the risetime.

Without going into the mathematics, an 11pF passive probe with a 6-inch ground lead will ring at about 140MHz when excited by a fast pulse. As the ring frequency increases, it tends to get outside the passband of the scope and is greatly attenuated. So to increase the ring frequency, use the shortest ground lead possible and use a probe with the lowest input C .

Probe Ground Lead Effects. The effect of inappropriate grounding methods can be demonstrated several ways. Figs. 1-16A, B and C show the effect of a 12-inch ground lead when used on various bandwidth scopes.

In Figure 1-16A, the display on the 15MHz scope looks OK because the ringing aberrations are beyond the passband of the instrument and are greatly attenuated. Figs. 1-16B and C show what the same signal looks like on 50MHz and 100MHz scopes.

Even with the shortest ground lead, the probe-DUT interface has the **potential** to ring. The potential to ring depends on the **speed** of the step function. The ability to **see** the resultant ringing oscillation depends on the scope system bandwidth.

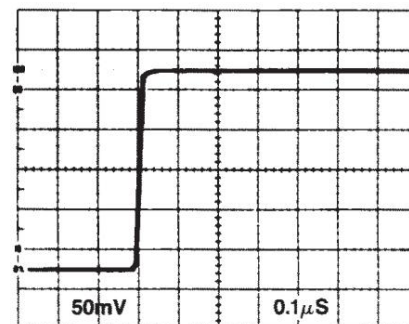


Figure 1-16A

Scope BW = 15MHz
Ground lead 12 inches

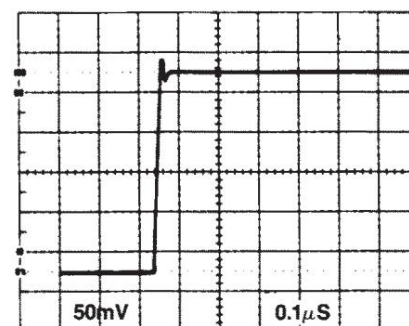


Figure 1-16B

Scope BW = 50MHz
Ground lead 12 inches

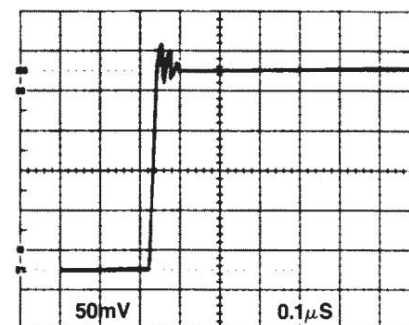


Figure 1-16C

Scope BW = 100MHz
Ground lead 12 inches

Figs. 1-17A through F show the effects of various grounding methods and ground lead lengths on the display of a very fast pulse. This is the most critical way of looking at ground lead effects: we used a fast pulse, with a risetime of about 70 pico seconds and a fast (400 MHz) scope with a matching P6137 probe.

Fig. 1-17A shows the input pulse under the most optimum conditions when using 50 ohm coax cable. Scope: the Tektronix 2465B with 50 ohm input and 50 ohm cable from a 50 ohm source. Displayed risetime is < 1 nSec.

Fig. 1-17B shows the same signal when using the scope-probe combination under the most optimum conditions. A BNC to probe adaptor or an in-circuit test jack provides a coaxial ground that surrounds the probe ground ring. This system provides the shortest probe ground connection available. Displayed risetime is < 1 nSec.

Figures 1-17C through E show the effects of longer ground leads on the displayed signal. Fig. 1-17C shows the effect of a short semi-flexible

ground connection, called a "Z" lead. Finally, Fig. 1-17F shows what happens when no probe ground lead is used.

How probe design affects your measurements

Probes are available in a variety of sizes, shapes and functions, but they do share several main features: a probe head, coaxial cable and either a compensation box or a termination.

The probe head contains the signal-sensing circuitry. This circuitry may be passive (such as a 9-M ohm resistor shunted by an 11 pF capacitor in a passive voltage probe or a 125-turn transformer secondary in a current probe); or active (such as a source follower or Hall generator) in a current probe or active voltage probe.

The coaxial cable couples the probe head output to the termination. Cable types vary with probe types.

The termination has two functions:

- to terminate the cable in its characteristic impedance.
- to match the input impedance of the scope.

The termination may be passive or active circuitry. For easy connection to various test points, many probes feature interchangeable tips and ground leads.

A unique feature of most Tektronix probes is the Tektronix-patented coaxial cable that has a resistance-wire center conductor. This distributed resistance suppresses ringing caused by impedance mismatches between the cable and its terminations when you're viewing fast pulses on wideband scopes.

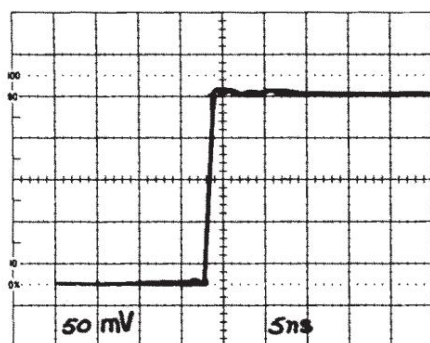


Figure 1-17A

50 ohm Source/Cable/2465B/50 ohm input

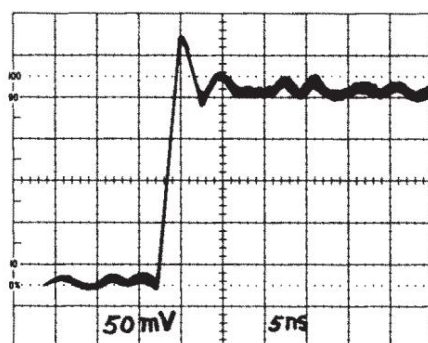


Figure 1-17C

P6137 - Probe/Z Ground Tr = 1.5 nS

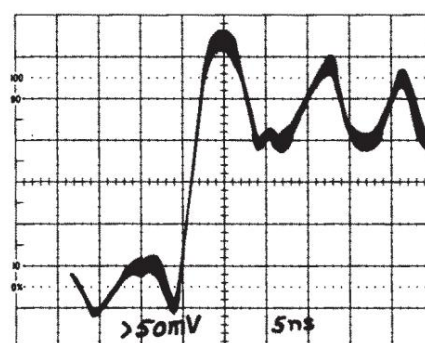


Figure 1-17E

P6137 - Probe/6" Gnd Lead Tr = 4 nS

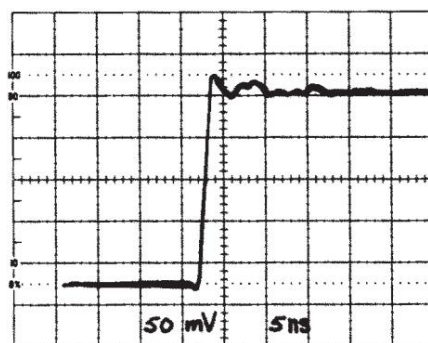


Figure 1-17B

P6137-BNC/Probe Adaptor Tr = < 1 nS

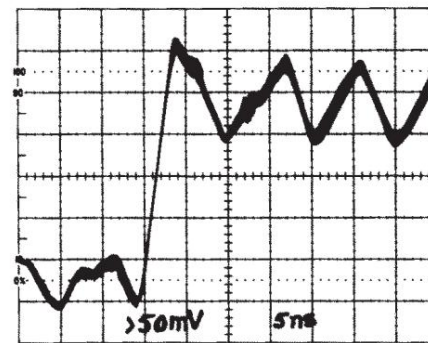


Figure 1-17D

P6137 - Probe/3" Gnd Lead Tr = 4 nS

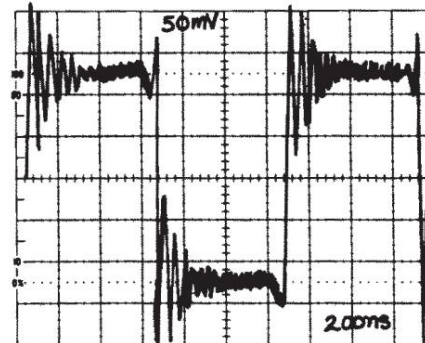


Figure 1-17F

No Ground Lead